MATH 202 Your name: _____

Learning target D1, version 1

Check whether a function is a solution to a differential equation, and explain why the function is or is not a solution.

Consider the differential equation

$$\frac{dh}{dt} = -2\sqrt{h}.$$

Check whether each of the following is or is not a solution.

1. $h(t) = t^2 + 4t + 4$ (Hint: Factor.)

2. $h(t) = (10 - t)^2$

3. $h(t) = 4e^{2t}$

MATH 202 Your name: _____

Learning target D1, version 2

Check whether a function is a solution to a differential equation, and explain why the function is or is not a solution.

Consider the differential equation

$$\frac{dT}{dt} = -k \cdot (T - 73),$$

where $k > 0$ is some constant. Check whether each of the following is or is not a solution:

(a) $T(t) = 73$

(b) $T(t) = 38e^{-kt}$

(c) $T(t) = 73 + 47e^{-kt}$